Roll No: | | | | | | | | |

SET 2

MID SEMESTER EXAMINATION Mar-2025

Name of The Program :M.Tech-Structural Engineering(MTSE)
Name OF the Course : Advance Concrete Design
Time: 90 Min

Semester : II
Course Code : MSE 202
Max Marks : 50

Note: i) Answer **all the questions** by **choosing any one of the sub questions**.
ii) Each question carries 10 Marks.

Q1 **Answer any one.** **(1X10=10 Marks)**

(a) Derive expression for maximum eccentricity based on limiting conditions of stresses? Explain about limiting zone of eccentricity for a prestressed concrete member? *(CO1)*

(OR)

(b) Explain the different target types (classes) of members we usually design in prestressed concrete design? *(CO1)*

Q2 **Answer any one.** **(1X10=10 Marks)**

(a) A post tensioned prestressed beam of rectangular section 250mm wide is to be designed for an imposed load of 12kN/m, UDL on a span of 12m. The stresses in concrete must not exceed 17N/mm^2 in compression and 1.4 n/mm^2 in tension at any time and the losses of prestress may be ssumed as 15% then calculate the minimum possible depth of the beam, and minimum prestressing force and corresponding eccentricity? *(CO1)*

(OR)

(b) Derive the expressions for minimum Pre Stressing Force required based on the limiting conditions of stresses in prestressed member? *(CO1)*

Q3 **Answer any one.** **(1X10=10 Marks)**

(a) State the assumptions made for analysis of secondary moments in continuous prestressed members? *(CO1)*

(OR)

(b) Design a simply supported type-2 prestressed beam with Mq=435kN-m, including an estimated Mg = 55kNm, Assume necessary data suitably).
The height of the beam is restricted to 920mm.
Prestress at transfer fp_0 = 1035N/mm^2. and the prestress at service fp_e=860N/mm^2 .
Based on the grade of the concrete the
Allowable compressive stresses are 12.5N/mm^2 at transfer and 11.0N/mm^2 at service.
Allowable tensile stresses are 2.1 N/mm^2 at transfer and 1.6 N/mm^2 at service.
7-wire strand with nominal diameter of 12.8 mm and nominal area of 99.3 mm^2 are used.? *(CO2)*

Q4 **Answer any one.** **(1X10=10 Marks)**

(a) Discriminate between the three general methods of continuous beam analysis to determine the secondary bending moments? *(CO2)*

(OR)

(b) Write an explanatory note on various advantages and disadvantages of continuous prestressed concrete structures? *(CO2)*

Q5 **Answer any one.** **(1X10=10 Marks)**

(a) State the assumptions made for analysis of secondary moments in continuous prestressed members?. Explain how the the secondary moments of a continuous prestressed member is analyzed using theorem of three moments? *(CO2)*

(OR)

(b) A prestressed concrete portal frame ABCD fixed at A and D has columns of 5m each and a Beam BC of 11m span. All the members have a cross section of 180mm wide and 200mm deep throughout. Columns are prestressed by a straight cable with an eccentricity of 70mm towards outside of the frame at B and C and Inner side of the frame at A and D. Transform BC is Prestressed by a Triangular cable with 50mm above the centroid at B and C and 80mm below the centroid at center of BC. Prestressing force in all the cables in 200kN. Calculate the secondary moments at A and B. Use consistent deformation method of analysis? *(CO2)*

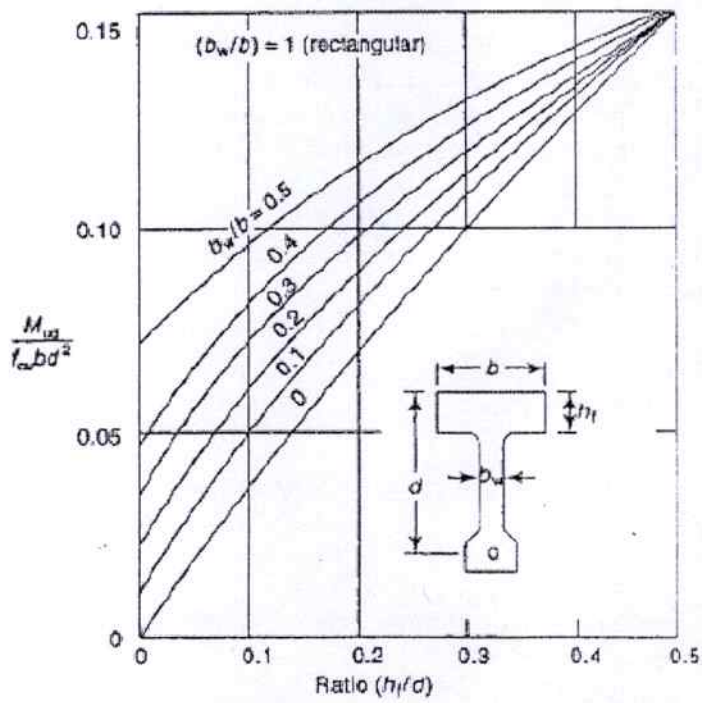

Chart for Preliminary Dimensioning of Prestressed Concrete Sections

Roll No:

SET 3